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EDUCATION AND SCHOLASTIC ACHIEVEMENTS

Program	Institute	% / CGPA	Year
B. Tech. (Ocean Engg. & IDDD Robotics)	Indian Institute of Technology, Madras	8.34	2027
STD XII	M.P. Board	86%	2022
STD X	C.B.S.E.	83.8%	2020

- Qualified for JEE Advance and secured a position in top 0.9% among 1.1 million participants conducted by IIT Bombay.
- Ranked among the top 0.4 percentile out of 9 lakh candidates nationwide in the competitive JEE Mains 2022 examination.
- Achieved a maximum rating of 1391 at Codeforces by constantly participating in competitive programming contests.

PROFESSIONAL EXPERIENCE

Analyst Internship at Deutsche Bank (May 26 – July 26)

- Engineered an **automated CI/CD vulnerability remediation** pipeline using **GitHub Actions** and **Python**, **reducing manual** enterprise **dependency updates** by **40%** and ensuring **100% validated build checks** prior to **PR generation**.
- Architected a fail-safe validation pipeline that dynamically **modifies package manager files** and **executes CI builds**, automatically **reverting failed upgrades** to preserve **continuous integration stability**.
- Deployed **Renovate Bot** across enterprise **Bitbucket repositories**, integrating **Artifactory and TeamCity** to **automatically validate builds** and **generate pull requests** for secure dependency updates.
- Developed a **custom Python parsing tool** for **Maven dependency trees**, integrating **OSV and NVD APIs** to automatically **extract CVEs**, evaluate **severity scores,** and **identify resolution versions** for both **direct and transitive vulnerabilities**.
- Awarded a **Pre-Placement Offer (PPO)** for architecting **scalable automation solutions** that can potentially transform enterprise vulnerability management and system security posture.

Cybersecurity Research Intern (May 24 – Aug 24)

R&D internship under Prof. NS Narayanaswamy

- **Simulated** advanced **credential harvesting** and **phishing attack vectors** using **Gophish and Evilginx** to **identify critical system vulnerabilities** and evaluate network **resilience against real-world scenarios**.
- Engineered **Proof of Concept (PoC) payloads** in controlled sandbox environments to demonstrate **initial access techniques,** delivering actionable threat intelligence for defensive system hardening.

PERSONAL PROJECTS

Latent Diffusion Model (LDM)

- Developed LDM from scratch using **PyTorch**, implemented **UNet** for noise prediction, **VAE** for latent space representation.
- Leveraged **Hugging Face** library to **tokenize 10,000-words** integrated **CLIP** for **prompt-based image generation.**
- Enabled controlled image generation by embedding prompts into **cross-attention** and aligning text with visuals precisely

Custom CNN Model for Card detection

- Achieved **91% accuracy** on custom **CNN model** by developing a 4-layer architecture for **reliable detection of rank & suit**
- **Optimized feed-forward time** by incorporating **batch normalization** layers and using OpenCV for live camera input feed
- **Reduced overfitting** by Applied **data augmentation** techniques, **L2 regularization,** and **dropout** for better generalization.

PID Controller

- Achieved precise **motion control** for **autonomous vessels** using **PID closed-loop controller** within the **ROS2** framework.
- Achieved **real-time position tracking** by utilizing **UWB** sensors and maneuvering equations to simulate vessel dynamics.
- Computed mass and Inertial matrices for accurate **motion prediction** and implemented them to enhance system stability.

Ace Bot

- Engineered a **robotic manipulator** using **CNN-based card detection**, enabling automated gameplay with human players.
- Engineered **forward** and **inverse kinematics** using **Jacobian inversion** for smooth and stable manipulator motion control.
- Optimized weight distribution and structural design by addressing challenges from motor weight, ensuring stable operation

POSITIONS OF RESPONSIBILITY

Envisage Core (Shaastra, CFI) (Apr 25 – Apr 26)

- Led **Team Envisage**, a multidisciplinary team comprising **Tech Ambience**, **Shows**, **GameDev**, and **XR** as verticals
- Represented a **70+ member team** across the **co-curricular sphere** and Oversaw **end-to-end operations of the team**
- Represented a 1**000+ member** organizing body as **Shaastra core** delivering key contributions across **India's largest student-run tech fest.**

Envisage Manager (Mar 24 – Mar 25)

- Managed a team of **20+ members**, overseeing project execution, logistics and coordination for technical deliverables.
- Worked on **Robotic manipulators** and implemented inverse kinematics using Jacobian inversion and developed custom CNN models

Envisage Coordinator (Jul 23– Mar 24)

- Coordinated with the team to develop Swift Canvas, advancing the previous year's **2D pen plotter** project.
- Showcased the project at **CFI Open House** 2024, **CFI Research Conclave** 2023, and **Shaastra** 2024, engaged over 9300.

COURSE WORK & SKILLS

OB3214: Deep learning for Computer Vision	EE6180: Advanced Topics in Artificial Intelligence
ID6040: Introduction to Robotics	MA2040: Probability Statistics and Stochastic Process
OB2202: Programming, Data Structures and Algorithms using Python	CS6847: Cloud Computing
ED5215: Introduction to motion planning	OE5011: Marine Robotics
EE6347: Devices and technologies for AI and neuromorphic computing	EE6412: Optimal Control

Programming Languages: Python, C, C++, SQL, Bash/Shell, MATLAB, Node.js
AI & Machine Learning: PyTorch, Hugging Face, Scikit-learn, Diffusion Transformers, Variational Autoencoders (VAE), DDPM, Computer Vision
Cloud, DevOps & HPC: Docker, Kubernetes (K8s), Helm, GitHub Actions, TeamCity, Bitbucket, Artifactory, High-Performance Computing (PBS)
Robotics & Simulation: ROS 2 (Jazzy), Gazebo, RViz2, SLAM, Arduino, Unity, Fusion 360, Gymnasium
Data & Web Frameworks: NumPy, Pandas, OpenCV, MediaPipe, Flask, Streamlit, KD-Trees, Linux (Arch)